Quade **Butler**

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Education

Doctor of Philosophy in Mechanical Engineering Dec 2025
McMaster University
Ontario Graduate Scholar, 2023 - 2024
Thesis: *Double Exponential Cubature Kalman Filtering: Theory and Applications*
Supervisor: *Dr. S. Andrew Gadsden*
GPA: *4.00/4.00*

Bachelor of Engineering in Mechanical Engineering 2020
University of Guelph
Minor in Mathematics
GPA: *3.80/4.00*

Experience

Research Assistant Jan 2026 – Present
ICE Lab | McMaster University Hamilton, ON

- Advanced doctoral research on nonlinear Bayesian state estimation by refining double exponential cubature methods for Gaussian-weighted integration and probabilistic estimation
- Developed robust filtering algorithms for nonlinear dynamic systems with measurement outliers, improving estimator accuracy, consistency, and numerical stability

PhD Researcher Jan 2021 – Dec 2025
ICE Lab | McMaster University Hamilton, ON

- Developed double exponential cubature for multivariate Gaussian-weighted integration, deriving deterministic cubature rules for probabilistic estimation in nonlinear dynamic systems
- Introduced the Double Exponential Cubature Kalman Filter (DECKF) extending the cubature method into a recursive Bayesian state estimator and benchmarking it against UKF, CKF, and Gauss-Hermite filtering baselines
- Developed and evaluated robust nonlinear filtering methods for state estimation under outliers and model uncertainty using Monte Carlo simulation, RMSE, NEES/NIS, and statistical consistency testing
- Built an interacting multiple model fault detection and operating regime classification workflow for nonlinear time-series data, achieving 99.2% accuracy and 99.1% macro-F1 across induced operating regimes

ICE Lab | Research collaboration with Ford Motor Company Livonia, MI

- Collaborated with Ford and university research teams on a production-feasible condition monitoring strategy for machine tool health monitoring and maintenance planning
- Designed an industrial machine tool monitoring framework by balancing diagnostic value, sensor constraints, deployment cost, and production feasibility
- Built Python pipelines for profiling, cleaning, synchronizing, and feature engineering noisy industrial datasets, reducing measurement and analysis time by 30x through workflow redesign and degradation modeling

Undergraduate Researcher May 2019 – May 2020
University of Guelph Guelph, ON

- Studied strain partitioning in advanced high-strength automotive steels using in-situ SEM tensile testing and micro digital image correlation
- Prepared specimens, processed SEM image sequences, optimized DIC parameters, and contributed to peer-reviewed publication in Materials Characterization

Publications

2025 **Gaussian Filtering Using a Spherical-Radial Double Exponential Cubature,** IEEE OJSP
Q. Butler, Y. Ziada, S. A. Gadsden

2024 **Strain partitioning characterization of advanced high strength steels using in-situ tensile tests with Micro digital image correlation – Methodology and analysis,** Materials
A. Bardelcik, Q. Butler

2023 **Preload Loss Detection in a Ball Screw System using Interacting Models,** IEEE OJIM
B. S. Sicard, Q. Butler, Y. Ziada, E. Hughey, S. A. Gadsden

2022 **Condition Monitoring of Machine Tool Feed Drives: A Review,** ASME JMSE
Q. Butler, Y. Ziada, D. Stephenson, S. A. Gadsden

Conferences

2025 **System identification and state estimation for magnetorheological dampers,** SPIE Signal Proc. XXXIV
P. Kosierb, Y. Wu, Q. Butler, B. Sicard, S. A. Gadsden

2025 **Friction Modeling and Monitoring for Machine Tool Health Management,** IEEE PHM
B. Sicard, Y. Wu, Q. Butler, S. A. Gadsden

2024 **Generating synthetic data for artificial intelligence solutions via a digital twin for condition monitoring in machine tools,** B. Sicard, Q. Butler, Y. Wu, S. Abdolahi, Y. Ziada S. A. Gadsden SPIE AI & ML II

2024 **Nonlinear Filtering Using the Double Exponential Transformation,** SPIE Signal Proc. XXXIII
Q. Butler, B. Sicard, W. Hilal, S. A. Gadsden, Y. Ziada

2023 **Design Considerations for Building an IoT Enabled Digital Twin Machine Tool Sub-System,** B. Sicard, Q. Butler, P. Kosierb, Y. Wu, Y. Ziada, S. A. Gadsden IEEE AIBThings

2023 **IIODT: Industrial Internet of Digital Twins for Hierarchical Asset Management in Manufacturing Sub-System,** B. Sicard, Q. Butler, P. Kosierb, Y. Wu, Y. Ziada, S. A. Gadsden IEEE AIBThings

2023 **Generalizing the Unscented Kalman Filter for State Estimation,** SPIE Signal Proc. XXXII
Q. Butler, W. Hilal, B. Sicard, Y. Ziada, S. A. Gadsden

2023 **Cognitive dynamic digital twin: enhancements for digital twin platforms based on human cognition,** B. Sicard, Q. Butler, Y. Ziada, S. A. Gadsden SPIE Big Data V

2023 **Experimental Setups for Linear Feed Drive Predictive Maintenance: A Review,** IEEE PHM
B. Sicard, Q. Butler, Y. Ziada, S. A. Gadsden

2022 **Rapid Parameter Estimation and Monitoring of CNC Feed Drive Systems,** SPIE Signal Proc. XXXI
Q. Butler, B. Sicard, E. Hughey, Y. Ziada, D. Stephenson, S. A. Gadsden

Teaching Assistantships

2024 **Mechanical Vibrations**, McMaster University
2024 **Engineering Design**, McMaster University
2022 **Mechatronics**, McMaster University
2022 **Engineering Measurements**, McMaster University
2020, 2021 **Differential Equations**, University of Guelph

Academic Services

2025 **Reviewer**, ASME Journal of Tribology